DR FENG CHEN

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ACADEMIC PROFILE

I am an interdisciplinary AI researcher with nine years of experience innovating **computer vision, deep learning, and AI agents** for **agriculture** and **healthcare**. I have published in leading conferences and journals (e.g., *Nature Communications* and *Pattern Recognition*). I have contributed to large BBSRC/EPSRC funded projects, and secured funding from Microsoft as Co-PI. I also have three years' teaching assistant experience and have supervised undergraduate, MSc, and PhD students.

PROFESSIONAL EXPERIENCE

PostDoc Research Associate, VIOS Collab, University of Edinburgh 02/2023 - present

- Working with: Prof Sotirios A. Tsaftaris
- Developed computer vision models and AI agents for plant phenotype analysis
  - Funded by BBSRC: PhenomUKRI The UK Plant and Crop Phenotyping Infrastructure (BB/Y512333/1, £2.4M)
- Developing vision-language models for AI-assisted surgery
  - Funded by EPSRC: Real-time Digital Twin Assisted Surgery (EP/X033686/1, £4M)

Teaching Assistant, University of Nottingham 09/2018 - 05/2021

- **Machine Learning**, Undergraduate/Postgraduate Module
- **Fundamental of Artificial Intelligence**, Undergraduate Module
- **Computer Vision**, Undergraduate/Postgraduate Module

EDUCATION

Ph.D. in Computer Science, Computer Vision Lab, University of Nottingham, U.K. 2018 - 2022

- Supervisors: Prof Andrew P. French, Prof Tony Pridmore
- Thesis: Learning with Fewer Labels in Deep Learning for Plant Phenotyping

M.Sc. in Signal Processing and Communications, University of Edinburgh, U.K. 2016 - 2017

- Supervisor: Prof Sotirios A. Tsaftaris
- Thesis: Citizen Scientists and Machine Learning to Help Feed the World (Best Thesis Award)

B.Eng. in Electronic and Information Engineering, Shantou University, China 2012 - 2016

Visiting Student, The Chinese University of Hong Kong (CUHK) 06 - 08/2015
Visiting Student, University of Calgary 09 - 12/2014

AWARDED GRANTS

Travel Scholarship, AIBIO-UK 3rd Annual Conference 2026

Co-PI, **Agricultural Foundation Models via Domain-Specific Pre-Training**, Microsoft Accelerate Foundation Models Research Grant (US$20,000) 09/2023 - 06/2025

SELECTED PUBLICATIONS

A Conversational Multi-Agent AI System for Automated Plant Phenotyping | Paper Link
Feng Chen, et al.
Nature Communications, 2026
Highlight: Developed with plant scientists and computer scientists, this work introduces a conversational system to automate plant phenotype analysis, covering phenotype measurement, data visualisation, and statistical analysis. It has led to a Best Poster Runner-Up Award at the AIBIO-UK Annual Conference and an internship project with Nottingham University; its Github repo has received 30 stars.

Stress Testing Concept Erasure with Large Language Model Agents | Paper Link
Yuyang Xue, **Feng Chen (co-first author)**, Zhihua Liu, Edward Moroshko, Steven McDonagh, Sotirios A. Tsaftaris
Under Review, 2026

Online OCF Diagnosis for DTP-PMSM via Semi-Supervised Learning and Compression-based Deep Learning Network with OOD Detection | Paper Link
Hao Yu, ..., Pericle Zanchetta, **Feng Chen**
IEEE Transactions on Transportation Electrification, 2026

Count2Density: Crowd Density Estimation without Location-level Annotations | Paper Link
Mattia Litrico, **Feng Chen**, Michael Pound, Sotirios A. Tsaftaris, Sebastiano Battiato, Mario Valerio Giuffrida
Pattern Recognition, 2026

Digital Twin-Assisted Surgery (DTAS): A Framework for Personalised and Predictive Care Across the Surgical Continuum (Book Chapter)
Lisa Asciak, ..., **Feng Chen**, ..., Grant D. Stewart, Wenmiao Shu
Digital Twins in Medicine: Architecture, Ethics and Clinical Applications for a Learning Health System, Springer, 2026

CSEval: A Framework for Evaluating Clinical Semantics in Text-to-Image Generation | Paper Link
Robert Cronshaw, Konstantinos Vilouras, Junyu Yan, Yuning Du, **Feng Chen**, Steven McDonagh, Sotirios A. Tsaftaris
International Symposium on Biomedical Imaging (ISBI), 2026

GMT: Guided Mask Transformer for Leaf Instance Segmentation (Selected as Oral Presentation) | Paper Link
Feng Chen, Sotirios A. Tsaftaris, Mario Valerio Giuffrida
Winter Conference on Applications of Computer Vision (WACV), 2025
Highlight: Developed a novel leaf segmentation model which performs well on both large and small leaves. WACV is a renowned computer vision conference (CORE A).

CRCE: Coreference-Retention Concept Erasure in Text-to-Image Diffusion Models | Paper Link
Yuyang Xue, Edward Moroshko, **Feng Chen**, Steven McDonagh, Sotirios A Tsaftaris
The British Machine Vision Conference (BMVC), 2025

SWiFT: Soft-Mask Weight Fine-tuning for Bias Mitigation | Paper Link
Junyu Yan, **Feng Chen**, Yuyang Xue, Yuning Du, Konstantinos Vilouras, Sotirios A. Tsaftaris, Steven McDonagh
Machine Learning for Biomedical Imaging (MELBA), 2025

Adapting Vision Foundation Models for Plant Phenotyping | Paper Link
Feng Chen, Mario Valerio Giuffrida, Sotirios A. Tsaftaris
International Conference on Computer Vision Workshops (ICCVW), 2023
Highlight: Applied parameter-efficient training to adapt large pre-trained vision models to plant phenotyping tasks under limited-data/computation settings. This work has received 40 citations since 2024.

Learning to Localise and Count with Incomplete Dot-annotations | Paper Link
Feng Chen, Michael Pound, Andrew P. French
ICCVW, 2021

Citizen Crowds and Experts: Observer Variability in Image-based Plant Phenotyping | Paper Link
Mario Valerio Giuffrida, **Feng Chen**, Hanno Scharr, and Sotirios A. Tsaftaris
Plant Methods, 2018

Towards Low-cost Image-based Plant Phenotyping Using Reduced-parameter CNN | Paper Link
John Atanbori, **Feng Chen**, Andrew P. French, Tony Pridmore
The British Machine Vision Conference Workshops (BMVCW), 2018

ACADEMIC SERVICES

Co-organiser, Computer Vision in Plant Phenotyping and Agriculture (CVPPA) Workshops, in conjunction with ICCV/ECCV

- **Communications Chair**: promote CVPPA through social media and community networks 2026
- **Paper Chair**: coordinate reviewer invitation, review process, and paper decision making 2025
- **Organising Committee Member** 2023, 2024

Young Scientists Editorial Board Member, Plant Phenomics Since 03/2025

Peer Reviewer for CVPR, ICCV, ECCV, BMVC, Applied Intelligence, Plant Phenomics, Plant Communications, NPJ Sustainable Agriculture, Earth System Science Data, Nature Scientific Data, Communications AI & Computing, Nature Astronomy

ACADEMIC TALKS

A Conversational Multi-Agent AI System for Automated Plant Phenotyping 06/2026
AIBIO-UK 3rd Annual Conference, University of Aberdeen

AI for Efficient and Accessible Plant Phenotyping (Keynote Presentation) 12/2024
PhenomUK & AIBIO-UK Joint Workshop, University of Nottingham

Accelerating Plant Phenotyping with AI 09/2024
UK Plant Phenomics Conference, University of Warwick

Retrieval-Augmented Generation: A Promising Approach to Enhance Fairness and Transparency in the Use of Large Language Models for Agriculture 05/2024
PhenoHarmonIS Workshop, Institut Agro, France

Citizen Scientists and Machine Learning to Help Feed the World 04/2018
Zooniverse Team, University of Oxford

STUDENT SUPERVISION

Vincent Nguyen, **Undergraduate Summer Intern**, University of Nottingham 06/2026–08/2026

- Project (Google DeepMind Research Ready Scheme): Agentic AI for Image-based Analysis

Junyu Yan, **PhD Student**, University of Edinburgh 09/2025–present

- Project: Automating medical observational study design with LLM agents

Yuyang Xue, **PhD Student**, University of Edinburgh 09/2025–present

- Project: Stress testing concept-erased generative models with LLM agents

Jingyu Sun, **Undergraduate Summer Intern**, University of Edinburgh 06/2024–09/2024

- Project: Pre-training agricultural foundation models with self-supervised learning and large-scale data

Ziyu Huang, **MSc Student**, University of Edinburgh 10/2023–09/2024

- Project (thesis): Active learning for medical image classification with less annotated data (awarded Distinction)

STUDENT AND ADMINISTRATION SERVICES

Residential Tutor, Broadgate Park, University of Nottingham 09/2020 - 08/2021

- Completed a 1-week training
- Provided pastoral support, helped maintain a safe residential environment, and resolved student and community complaints in a university hall with 2,000 undergraduate and postgraduate students

Global Buddies Mentor, University of Nottingham 09/2019 - 12/2019

- Completed a 1-week training
- Coordinated social activities for international students to adapt to life in Nottingham

SELECTED AWARDS

Best Poster Prize Runner Up (out of 22 posters), AIBIO-UK 3rd Annual Conference in Aberdeen 2026

BMVC Outstanding Reviewer, British Machine Vision Association (BMVA) 2024

International PhD Student Full Scholarship, University of Nottingham 2018 - 2021

Best Master Thesis Award, School of Engineering, University of Edinburgh 11/2017

National Scholarship (top 0.2% nationally), Shantou University 06/2016

Nomination for the Shantou University Medal (the highest honor for graduates), Shantou University 06/2016

First-class Scholarship for Overseas Study (for MSc), Li Ka Shing Foundation/Shantou University 06/2016